

GLMs for Applied Research

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Session 6: Overdispersion & Solutions

Topics we will cover in this session

- Poisson assumptions & overdispersion
- Quasi Poisson models
- Negative Binomial Models
- Choosing between QP and NB models

$$\text{Var}(Y_i) > E(Y_i)$$

The variance in the data exceeds what the Poisson distribution allows — it is not a modeling choice, it is a property of the data itself.

POISSON ASSUMES

Mean = Variance

Spread grows only as fast as the average count — one parameter controls both.

REAL DATA SHOWS

Variance $\gg$ Mean

Extra variability from unobserved factors inflates the spread.

Causes of overdispersion

- **Unobserved heterogeneity**

Individuals share a covariate profile but differ on unmeasured traits that also drive the count.

- **Excess zeros**

More zero counts occur than the Poisson distribution predicts, widening the spread.

- **Clustering & correlation**

Observations within a group (sites, time periods) are not independent, inflating variance.

- **Omitted covariates**

A relevant predictor is left out of the model, so its effect shows up as extra unexplained variance.

Consequences of Ignoring Overdispersion

Fitting an ordinary Poisson model to overdispersed data doesn't bias the coefficients much — but it badly understates their uncertainty.

NAIVE POISSON FIT

- Standard errors underestimated
- Confidence intervals too narrow
- Wald / z-test statistics inflated
- p-values misleadingly small
- Spurious “significant” effects

DISPERSION-ADJUSTED FIT

- Standard errors correctly scaled
- Confidence intervals reflect true uncertainty
- Test statistics properly calibrated
- p-values trustworthy
- Fewer false discoveries

Detecting overdispersion

$$\hat{\phi} = \sum (\text{Pearson residuals})^2 / (n - p)$$

$n - p$ = residual degrees of freedom

Rule of thumb: $\hat{\phi} \approx 1 \rightarrow$ no overdispersion

$\hat{\phi} > 1.5-2 \rightarrow$ investigate further

```
model_pois <- glm(y ~ x1 + x2,  
  family = poisson(link = "log"), data = df)  
  
phi_hat <- sum(residuals(model_pois,  
  type = "pearson")^2) / model_pois$df.residual  
phi_hat
```

- A dispersion statistic well above 1 signals that Poisson's variance assumption is violated
- Also inspect the ratio of residual deviance to degrees of freedom — the same logic applies
- R's `summary.glm()` reports this deviance/df ratio automatically for Poisson fits
- Formal tests exist too (e.g. `AER::dispersiontest()`), but the quick statistic above is usually enough to decide whether action is needed

Overdispersion: Solutions

- Quasi-Poisson: inflates standard errors by $\sqrt{\phi}$ — same coefficients, correct inference
- Negative Binomial regression: adds an individual-level random effect; models $\text{Var}(Y) = \mu + \mu^2/\theta$ (preferred approach)
- Zero-inflated models: when zeros are more frequent than Poisson predicts

Quasi-Poisson Theory

$$\log(\mu_i) = \beta_0 + \beta_1 x_{i1} + \dots$$

same mean model as Poisson

$$\text{Var}(Y_i) = \varphi \cdot \mu_i$$

φ is a free dispersion parameter, no longer fixed at 1

- φ is estimated directly from the Pearson residuals, not assumed in advance
- Not a full probability distribution — built from quasi-likelihood, using only a mean and variance relationship

WHAT CHANGES — AND WHAT DOESN'T

- Point estimates (β) are identical to a standard Poisson fit — the mean model hasn't changed
- Standard errors are inflated by a factor of $\sqrt{\varphi}$, giving honest uncertainty
- z-statistics and p-values shrink accordingly — fewer false positives
- No true likelihood exists, so AIC / BIC are not defined for this model
- Best suited when the variance grows roughly linearly with the mean

Fitting Quasi-Poisson model

```
model_qp <- glm(  
  y ~ x1 + x2,  
  family = quasipoisson(link = "log"),  
  data = df  
)  
  
summary(model_qp)  
  
# extract the estimated dispersion  
disp <- summary(model_qp)$dispersion  
disp  
  
# coefficients match Poisson; SEs scale by sqrt(disp)  
coef(model_qp); summary(model_qp)$coefficients
```

READING THE OUTPUT

- `summary()` prints “Dispersion parameter for quasipoisson family taken to be ϕ ”
- Compare that value to 1 — far above it confirms overdispersion
- Coefficient table's SEs are already the corrected ones
- No `$aic` field is returned — model comparison needs a different tool (more on this later)

Negative Binomial Theory

A POISSON–GAMMA MIXTURE

$$\lambda_i \sim \text{Gamma}(\mu_i, \theta)$$

↓ *random effect absorbs unobserved heterogeneity*

$$Y_i \mid \lambda_i \sim \text{Poisson}(\lambda_i)$$

↓ *integrate out λ_i*

$$Y_i \sim \text{NegBinom}(\mu_i, \theta)$$

$$\text{Var}(Y_i) = \mu_i + \mu_i^2 / \theta$$

θ is estimated from the data — smaller θ means more overdispersion

As $\theta \rightarrow \infty$, this reduces exactly to the Poisson variance μ_i

- Variance grows quadratically in μ — a different shape than quasi-Poisson's linear $\phi\mu$
- A genuine probability distribution with its own likelihood — estimated by full maximum likelihood
- Because it's a proper likelihood, AIC, BIC, and likelihood-ratio comparisons are all valid

Fitting Negative Binomial

```
library(MASS)

model_nb <- glm.nb(y ~ x1 + x2, data = df)
summary(model_nb)

# estimated dispersion parameter theta
model_nb$theta
model_nb$SE.theta

# proper likelihood -> AIC is valid
AIC(model_nb)

# formal test: Poisson vs. Negative Binomial
library(pscl)
odTest(model_nb)
```

READING THE OUTPUT

- theta is printed near the bottom of summary() with its own standard error
- A small theta (relative to its SE) means strong overdispersion
- Coefficients can differ slightly from Poisson, since the whole likelihood is refit
- odTest() from pscl formally tests Poisson ($\theta \rightarrow \infty$) against Negative Binomial

Quasi-Poisson vs Negative Binomial

	Quasi-Poisson	Negative Binomial
Basis	Quasi-likelihood (mean-variance rule only)	Full probability distribution
Variance function	$\text{Var}(Y) = \phi\mu$ — linear in μ	$\text{Var}(Y) = \mu + \mu^2/\theta$ — quadratic in μ
Coefficient estimates	Identical to Poisson MLE	Re-estimated; can differ slightly
Standard errors	Inflated by $\sqrt{\phi}$	From the NB likelihood directly
AIC / BIC available	No — no true likelihood	Yes — valid model comparison
R function	<code>glm(family = quasipoisson)</code>	<code>MASS::glm.nb()</code>

Best model to go with

Fit a Poisson GLM

Start with the simplest model as a baseline

Compute $\hat{\phi}$

Pearson χ^2 / residual df

$\hat{\phi} \approx 1$?

Poisson is adequate — stop here

Fit QP & NB

If $\hat{\phi}$ is large, fit both alternatives

Compare

AIC (NB), theory fit, and diagnostics

Rule of thumb: reach for quasi-Poisson when you only need corrected standard errors; reach for negative binomial when you want a full likelihood, valid AIC comparisons, or you suspect the variance grows faster than the mean.

Recap & What's next

- Ignoring it doesn't bias coefficients much, but it makes standard errors and p-values untrustworthy.
 - Quasi-Poisson keeps the Poisson mean model and simply rescales the variance by an estimated ϕ .
 - Negative Binomial models the extra variance explicitly through θ , giving a full likelihood and valid AIC.
 - Check $\hat{\phi}$ first, then let the goal — corrected SEs vs. a complete generative model — guide your choice.
- **Next:** Hands-on - QuasiPoisson and Negative Binomial